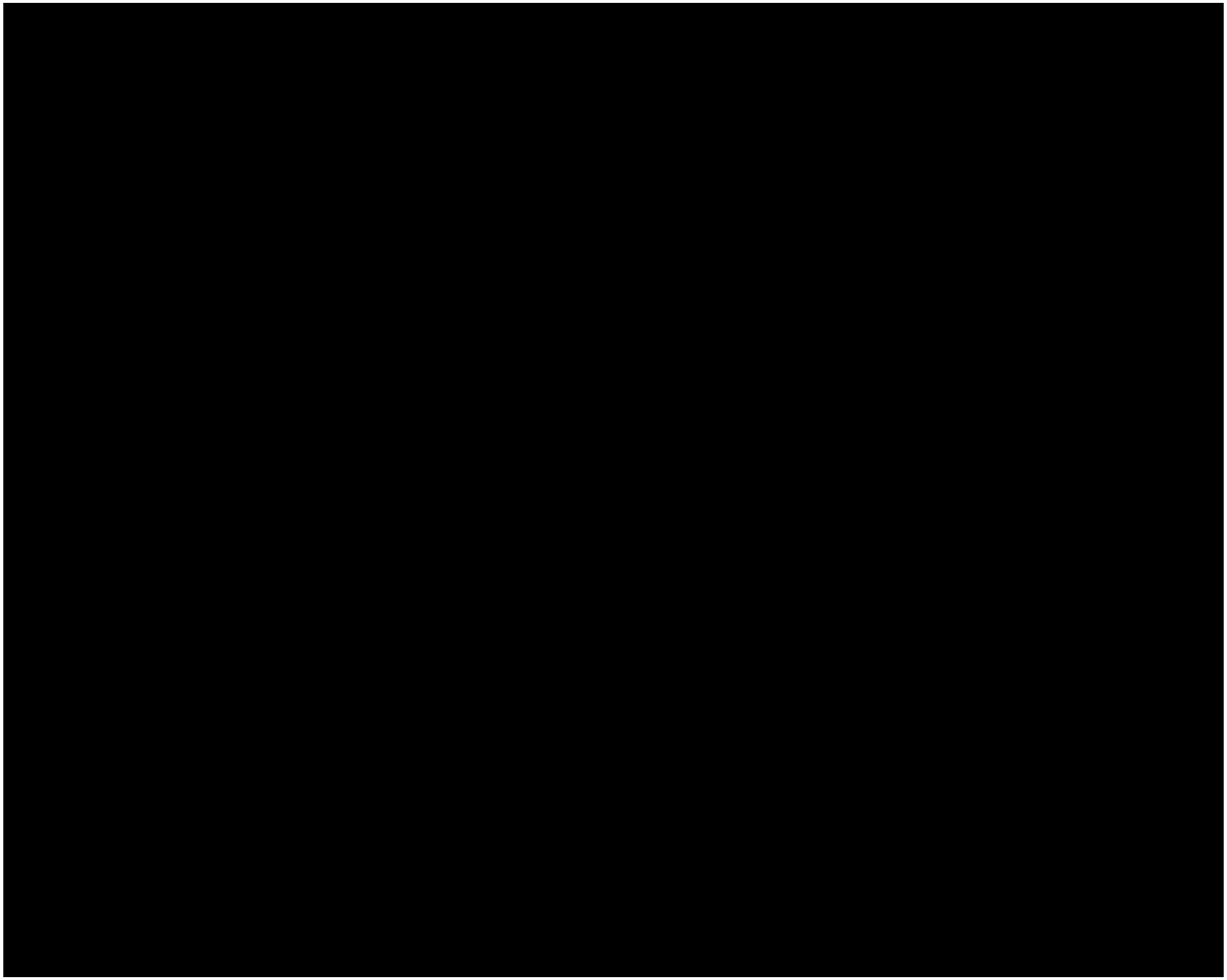
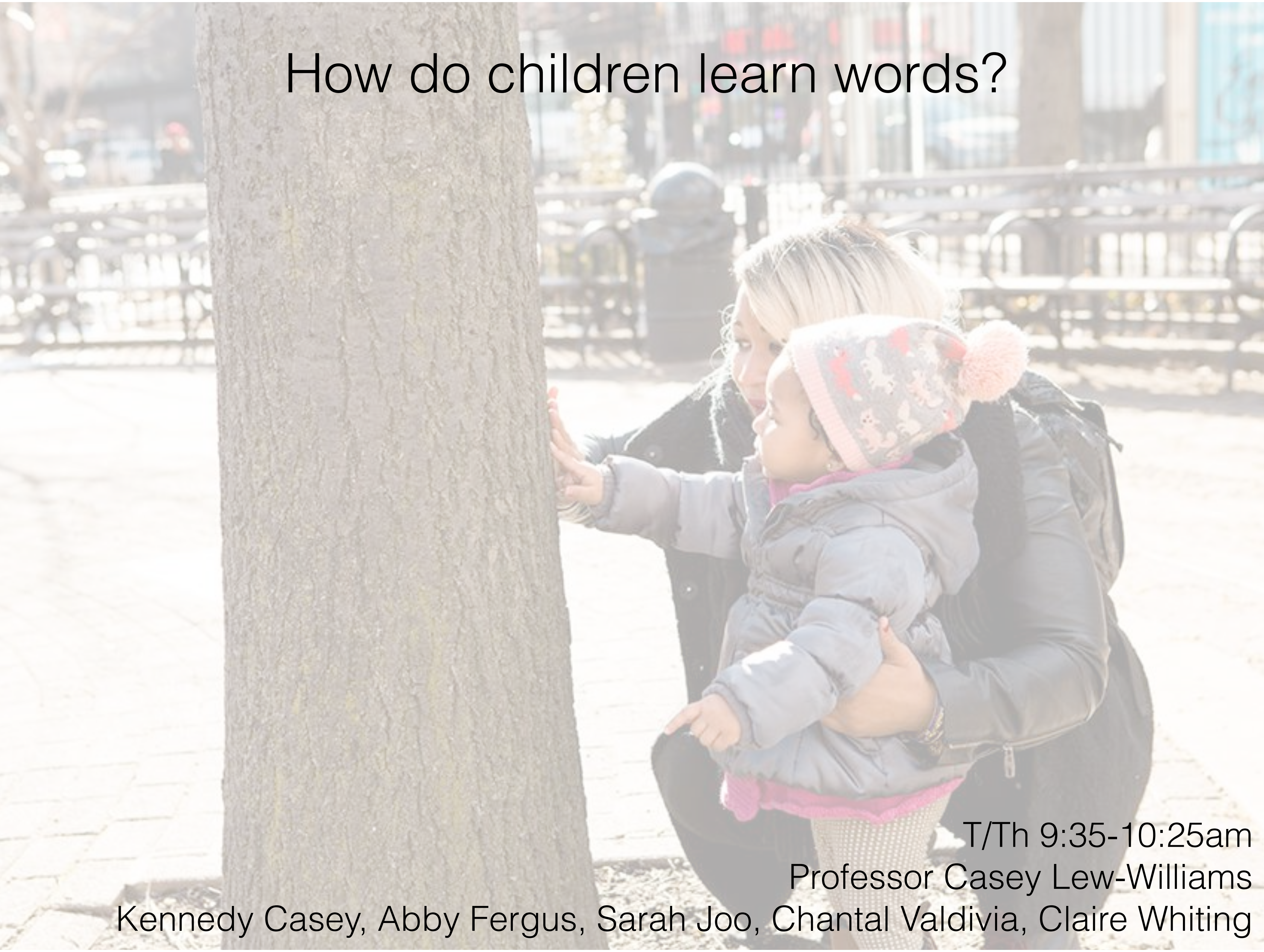
# A word segmentation exercise: Silbo Gomero



http://www.youtube.com/watch?v=AqdAnGDMU2k

# How do children learn words?



T/Th 9:35-10:25am

Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

James while John had had had had had had had had had had had a better effect on the teacher.

James, while John had had *had*, had had *had had; had had* had had a better effect on the teacher.

https://en.wikipedia.org/wiki/James_while_John_had_had_had_had_had_had_had_had_had_had_had_a_better_effect_on_the_teacher

# Journal article outline

- An **outline** of your journal article assignment is due before your precept next week.

- Your preceptor will give you feedback. So, you will benefit if you provide a clear articulation of your research question, hypothesis, and study design. Bullet points are fine!

- You <u>may not</u> use AI in any way, not even to brainstorm. See wording about my strict AI ban in the syllabus. We are here to provide a Princeton education, and generating work yourself is the foundation of it. Even supplemental AI use undermines a Princeton education.

- See details about the assignment on Canvas / Assignments / Journal Article Assignment

How do children learn new words?

# Quine's 'gavagai' problem

- Word learning seems straightforward: child sees something and then hears a word…

- But, Quine's "gavagai" problem

Rabbit

Ears

Fast

Running

Dinner

Look

Undetached rabbit parts

ooh a visuospatial context


NO BUSKING PLEASE
DURING SCHOOL HOURS
9am to 5pm
John Bull
CITY
OF
YORK


Welcome to York
6
MINT VELVET
AFTERNOON INDULGENCE
Thank You for supporting *KENYAN ORPHANS
Donations Gratefully Accepted
Denis is Busking for *KENYAN ORPHANS
Thank you for your support


Weather
12
21
anfairpwllgwyngyllgogerychwyrndrobwllllantysiliogogogoch
Earlier today

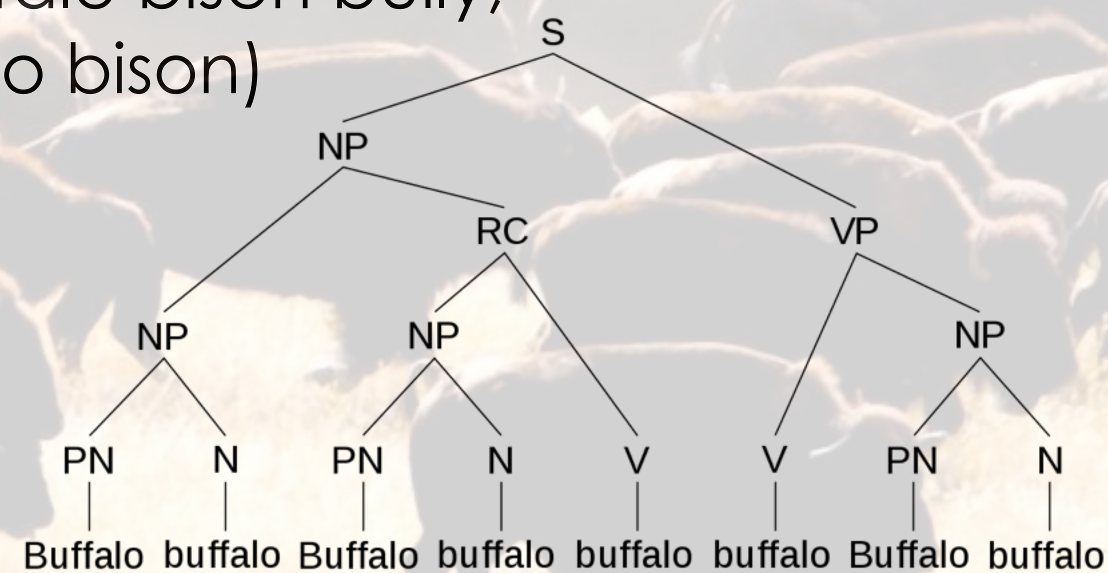
## This is a ship-shipping ship, shipping shipping ships

**Buffalo buffalo buffalo Buffalo buffalo.**

(bison from Buffalo bully other bison from Buffalo)

**Buffalo buffalo Buffalo buffalo buffalo buffalo Buffalo buffalo.**

(Buffalo bison who other Buffalo bison bully, themselves bully other Buffalo bison)



# Very early word comprehension and production

- Babies respond to their own name around 4 months of age.
- Between 2 and 5 months of age, infants learn that their non-cry vocalizations get responses from caregivers.

(Mandel, Pisono, & Jusczyk, 1995; Elmlinger et al., 2022)

# Babbling

- Provides practice with articulating relevant sounds.

- Babbling actively changes the type of input that infants receive from caregivers.
    - Caregivers are more likely to respond to babbling and to adult-like vowels
    - Contingent speech from caregivers is more sensitive and imitative
    - **So, by babbling, infants create social interactions that facilitate their own language learning.**

# Babbling

Deaf infants babble with their hands around the same time as hearing infants!
(Pettito et al., 2001)

Often babbling and first words coexist…

http://video.google.com/videoplay?docid=8832999562201431834#

# Babbling

- Babbling at an object is a signal of infants' interest, offering a good moment to hear words from parents.


(Goldstein & Schwade, 2009)

# Physiology shapes infants' vocalizations

- Infants time their vocalizations to fluctuations in heart rate, which helps them produce louder and more recognizable words.

(Borjon, Sahoo, et al., 2025)

# First words

Table 4
*Rank-Ordered Top 20 Words for Children Who Can Say 1–10 Words on CDI and Percentage of Children Producing Them, by Language*

| United States ($n$ = 264) | Hong Kong ($n$ = 367) | Beijing ($n$ = 336) |
|---|---|---|
| **Daddy (54)** | **Daddy (54)** | **Mommy (87)** |
| **Mommy (50)** | *Aah (60)* | **Daddy (85)** |
| BaaBaa (33) | **Mommy (57)** | *Grandma—Paternal (40)* |
| **Bye (25)** | *YumYum (36)* | *Grandpa—Paternal (17)* |
| **Hi (24)** | *Sister—Older (21)* | **Hello?/Wei? (14)** |
| **UhOh (20)** | **UhOh (Aiyou) (20)** | *Hit (12)* |
| Grr (16) | *Hit (18)* | Uncle—Paternal (11) |
| Bottle (13) | **Hello?/Wei? (13)** | Grab/Grasp (9) |
| *YumYum (13)* | Milk (13) | *Auntie—Maternal (8)* |
| Dog (12) | Naughty (8) | **Bye (8)** |
| No (12) | *Brother—Older (7)* | **UhOh (Aiyou) (7)** |
| **WoofWoof (11)** | *Grandma—Maternal (6)* | *Ya/Wow (7)* |
| *Vroom (11)* | *Grandma—Paternal (6)* | *Sister—Older (7)* |
| Kitty (10) | **Bye (5)** | **WoofWoof (7)** |
| *Ball (10)* | Bread (5) | *Brother—Older (6)* |
| Baby (7) | *Auntie—Maternal (4)* | Hug/Hold (6) |
| Duck (6) | *Ball (4)* | Light (4) |
| Cat (5) | *Grandpa—Paternal (4)* | *Grandma—Maternal (3)* |
| Ouch (5) | Car (3) | Egg (3) |
| Banana (3) | **WoofWoof (2)** | *Vroom (3)* |

First words can be objects, but many are *not*.

Instead, they're social and routine-based words:

uh-oh, hi, bye-bye, yum, more, wow, shh, all-gone, yes, no, thank-you, up, night-night

(Casey, Potter, Lew-Williams, & Wojcik, 2023; Tardif et al., 2008)

# First multi-word combinations

My kids:

Beer dada.

See pee.

# Underextensions and Overextensions

**Underextensions**: ‘doggie’ only refers to one dog (e.g., Fido).

**Overextensions**: one word refers to many things.

**ball**: ball; balloon; marble; apple; egg; pom-pom; spherical water tank (Rescorla, 1980)

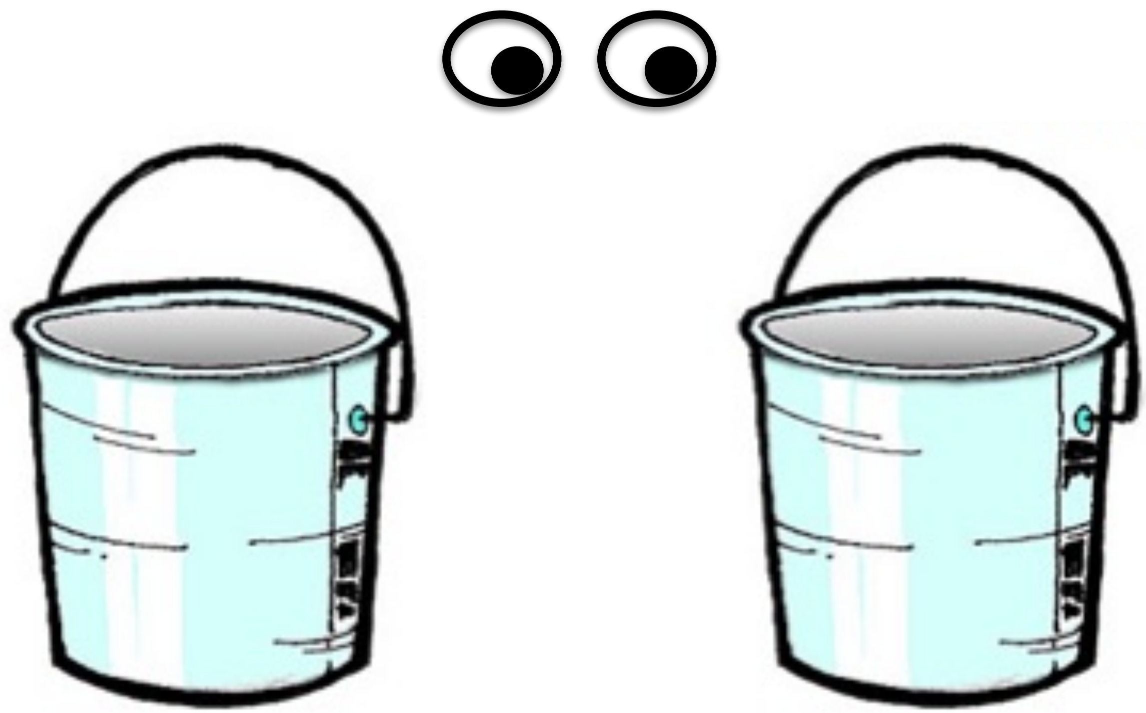
**cat**: cat; cat’s usual location on top of TV when absent (Bowerman, 1978)

**snow**: snow; white tail on a horse; white flannel bed pad; white puddle of milk on the floor (Bowerman, 1978)

# How do young children learn object-label mappings?

- 18-month-olds can use <u>social cues</u>, such as an adult's eye gaze, to map novel words onto novel objects

*There's a modi in there!*

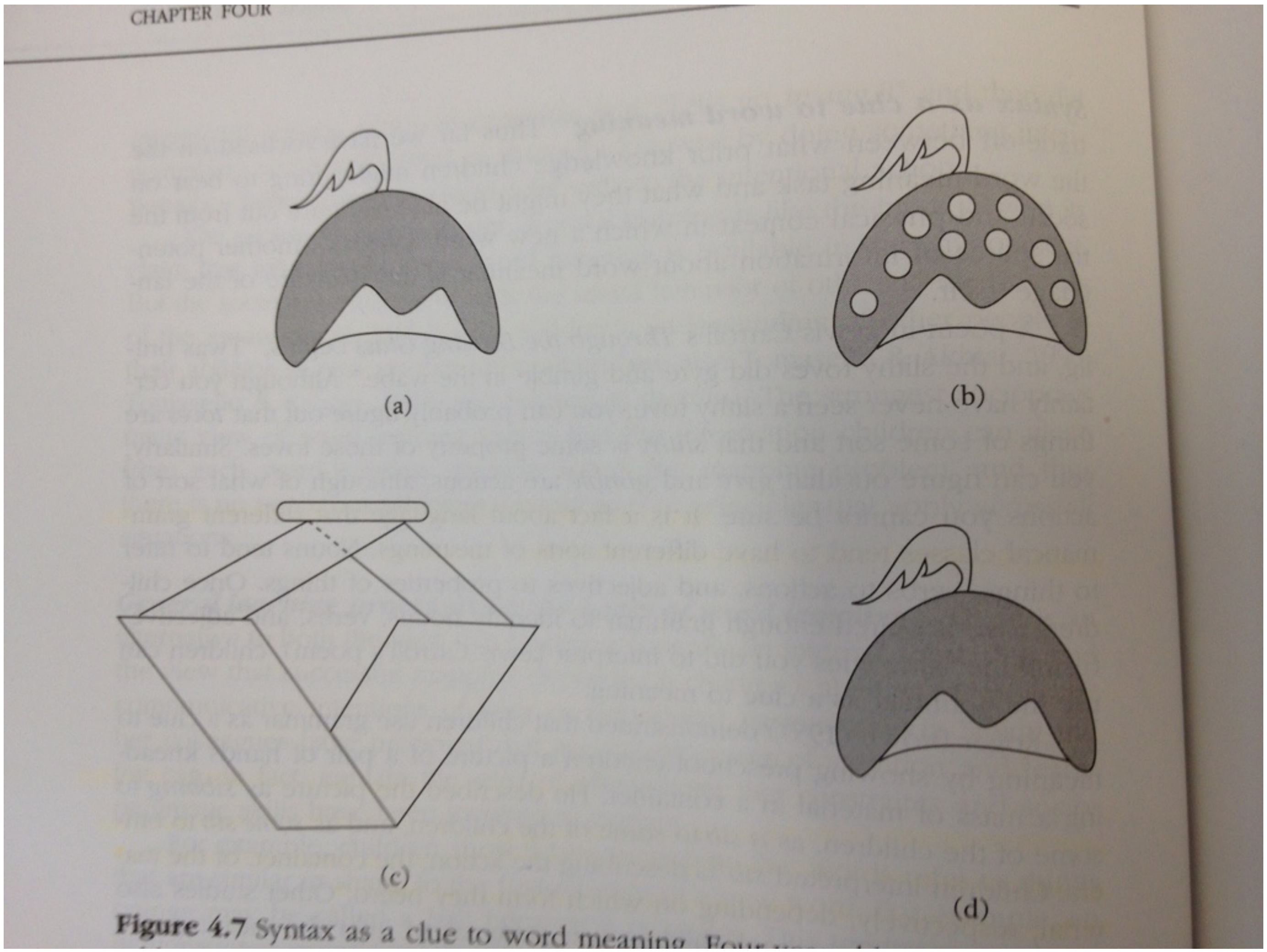


# How do young children learn object-label mappings?

- Children can use emotion to do this too

# How do young children learn object-label mappings?

- Children can use the <u>grammar</u> of a sentence to distinguish between words that contrast members of the same category, and words that do not.




Find the fep.

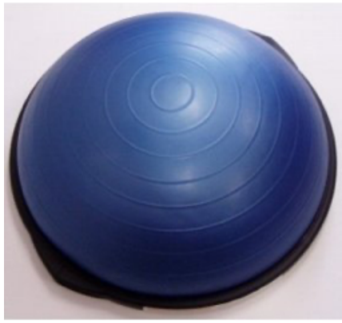
Now find the fep one.

- Children can use logic to learn words. Given the infinite possible hypotheses for word meaning, maybe logic can help restrict the possibilities…

**Whole Object**

"Look at the toma"

**Mutual Exclusivity**



"Go in the garage and get me the bosu"

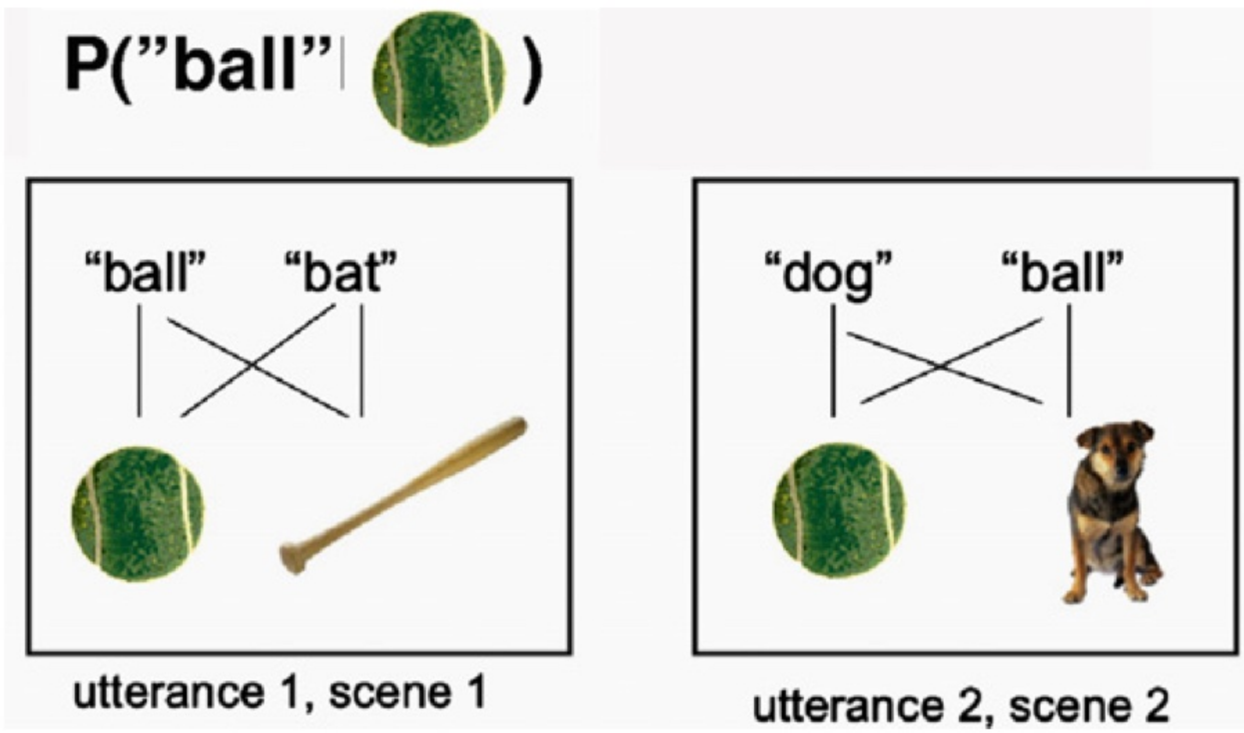
1. Flim moge poe rungmawt

2. Flim rungmawt blergenfall kaw

3. Flim bliffen poe blergenfall

# Cross-situational word learning

- You just learned words even in a highly ambiguous learning context!
- "Cross-situational word learning" (Yu & Smith, 2007)




Adults, and even 12-14-month-olds
can track complicated object-label links over time


SERVICE & PRICE CHECK
Scoop Away
Scoop Away
Fresh Step
Fresh Step
TIDY CATS
Deli-Cat

But toddlers have short arms.

Short arms, as it turns out, are stable organizing principles for toddler experience. Combined with infants' curiosity, they have many useful experiences to learn words.

Babies systematically create situations where one object is visually large, which reduces clutter in the world.

Naturalistic data show that infants almost always experience naming events at two extremes:

Tons of moments with extremely clear reference, tons of moments with extremely ambiguous reference, and not much in between.

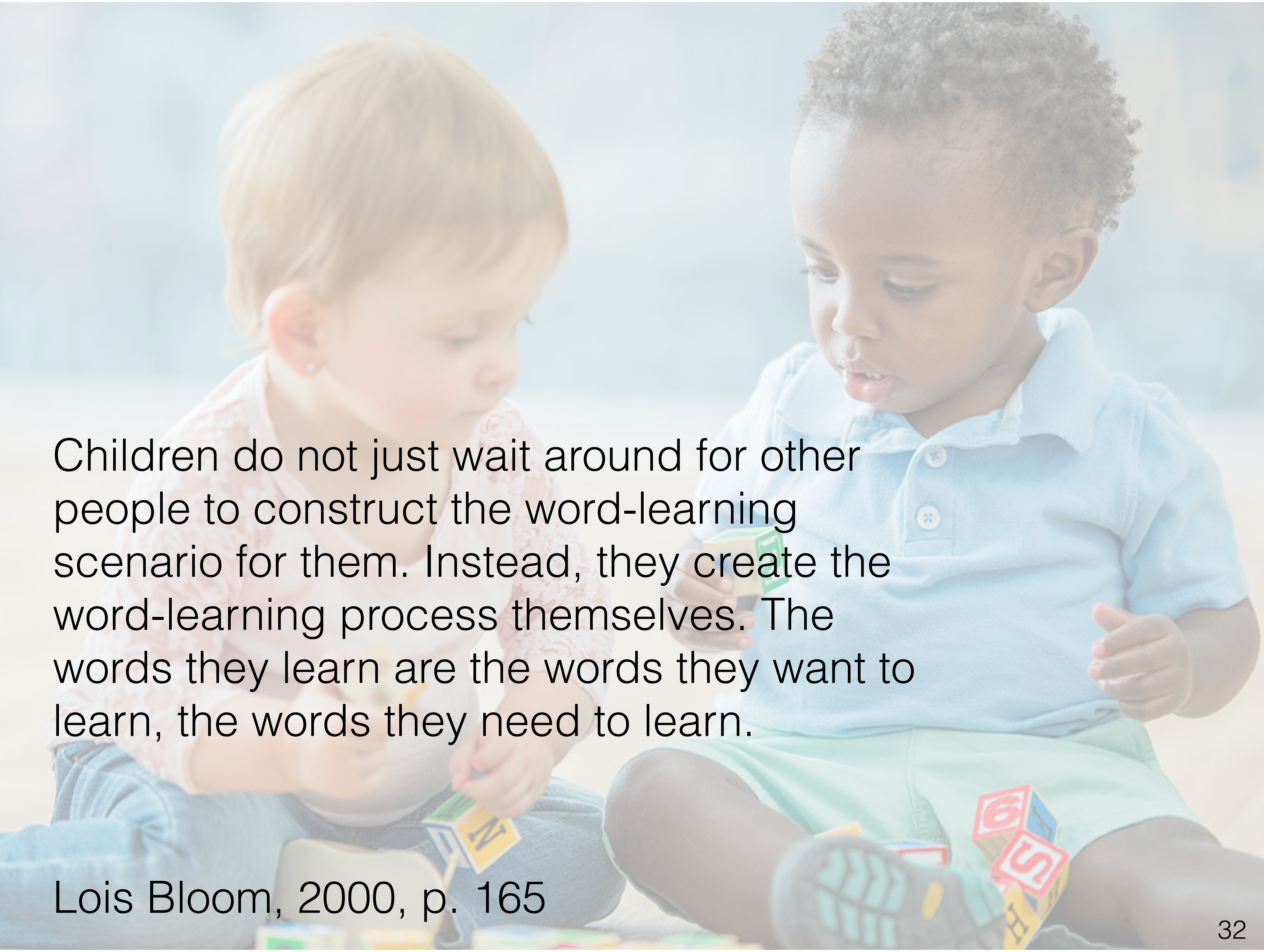
Somehow they learn over time from this combination. Nobody knows how.

(Yu, Smith, et al.)

# Children learn new words best for categories <u>they're</u> interested in!

| Animals | Clothes | Vehicles | Drinks |
|---|---|---|---|
| Broad | | Narrow | |
| Pango | Gugel | Rikscha | Malpe |
|  |  |  |  |

(Ackerman, Hepach, & Mani, 2020)



Children do not just wait around for other people to construct the word-learning scenario for them. Instead, they create the word-learning process themselves. The words they learn are the words they want to learn, the words they need to learn.

Lois Bloom, 2000, p. 165

In a study of 3,387 children in Canada, vocabulary size at age 2 predict school-related social skills at age 8.

(Wieczorek, Hentges, Tough, & Graham, 2025)

How do children process language in real time?

# The Lion-Eating Poet in the Stone Den

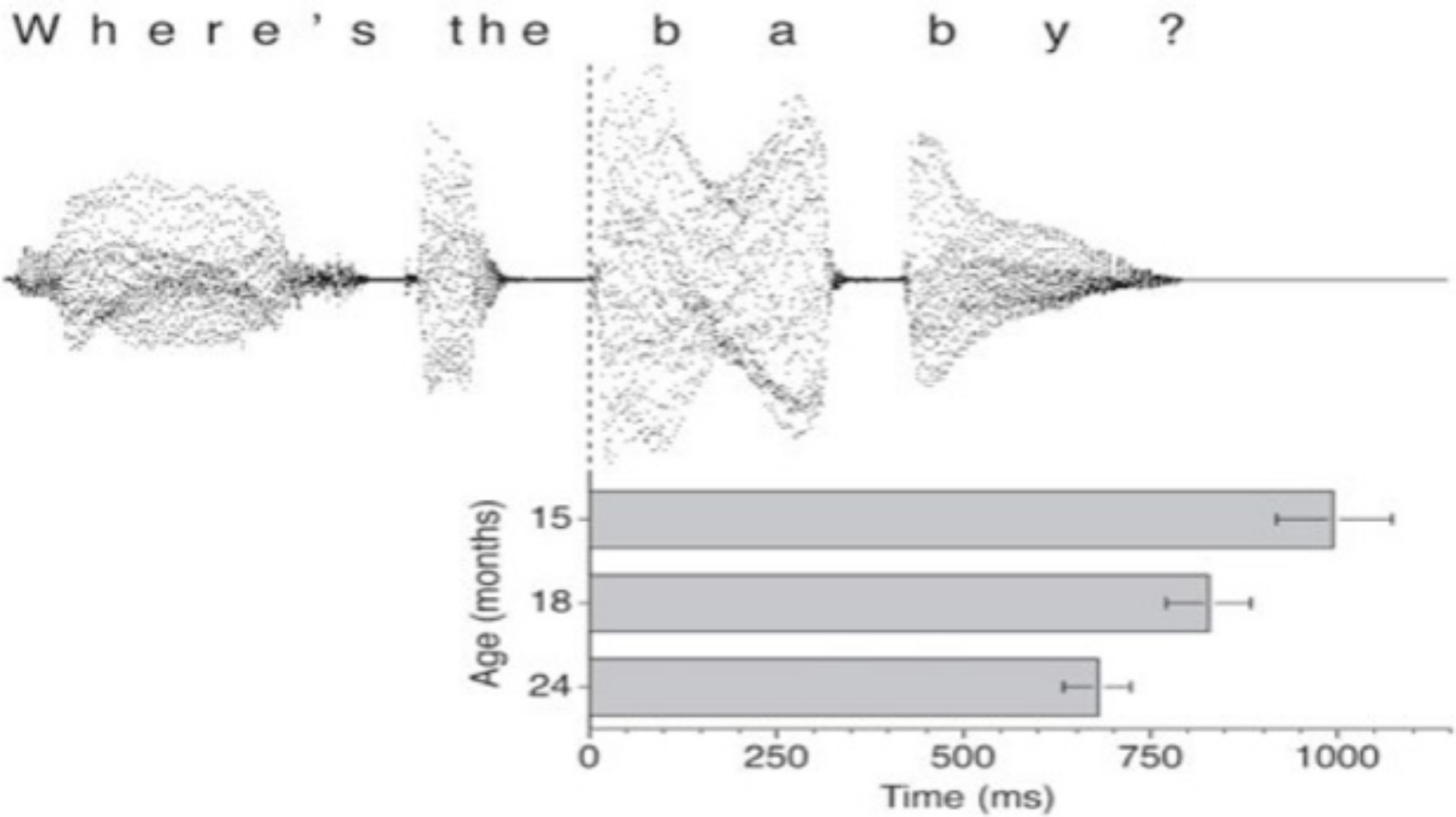
shi2 shi4 shi1 shi4 shi1 shi4 shi4 shi1. shi4 shi2 shi2 shi1. shi4 shi2 shi2 shi4 shi4 shi4 shi1 shi2 shi2. shi4 shi2 shi1 shi4 shi4 shi4 shi2. shi4 shi1 shi4 shi4 shi4 shi4 shi4 shi4 shi2 shi1. shi4 shi3 shi4 shi3 shi4 shi2 shi1 shi4 shi4 shi4 shi2 shi4 shi2 shi1shi1. shi4 shi2 shi4 shi2 shi4 shi1. shi4 shi3 shi4 shi4 shi2 shi4 shi2 shi4 shi4. shi4 shi3 shi4 shi2 shi4 shi2 shi1 shi1 shi2 shi2. shi3 shi4 shi3 shi2 shi1 shi1 shi2 shi2 shi2 shi1 shi1 shi4 shi4 shi4 shi4

A poet by the name of Shih Shih living in a stone den was fond of lions. As he had taken an oath to eat ten lions, he went out to the market every day at ten o'clock in order to look for lions. It was at the time when all of a sudden ten lions came to the market and also Shih Shih went to the market at once realizing these ten lions. Relying on his (bow and) arrows, he caused these ten lions to pass away. Shih picked up the corpses of these ten lions, and as he went to the stone den, the stone chamber was damp. Shih had the stone den wiped by his servant. As the stone den was cleaned, it was the time that Shih began trying to eat the meal of these ten lions' corpses and he began to realize that these ten dead lions in fact were ten stone lions' corpses and he tried to get rid of this matter.

石室诗士施氏，嗜狮，誓食十狮。适施氏时时适市视狮。十时，适十狮适市。是时，适施氏适市。氏视是十狮，恃矢势，使是十狮逝世。氏拾是十狮尸，适石室。石室湿，氏使侍拭石室。石室拭，氏始试食是十狮尸。食时，始识是十狮尸，实十石狮尸。示释是事。

Word learning isn't just about labeling stuff. It's about quickly processing everything that comes your way…

# Developmental gains in speed of spoken word recognition over 2nd year

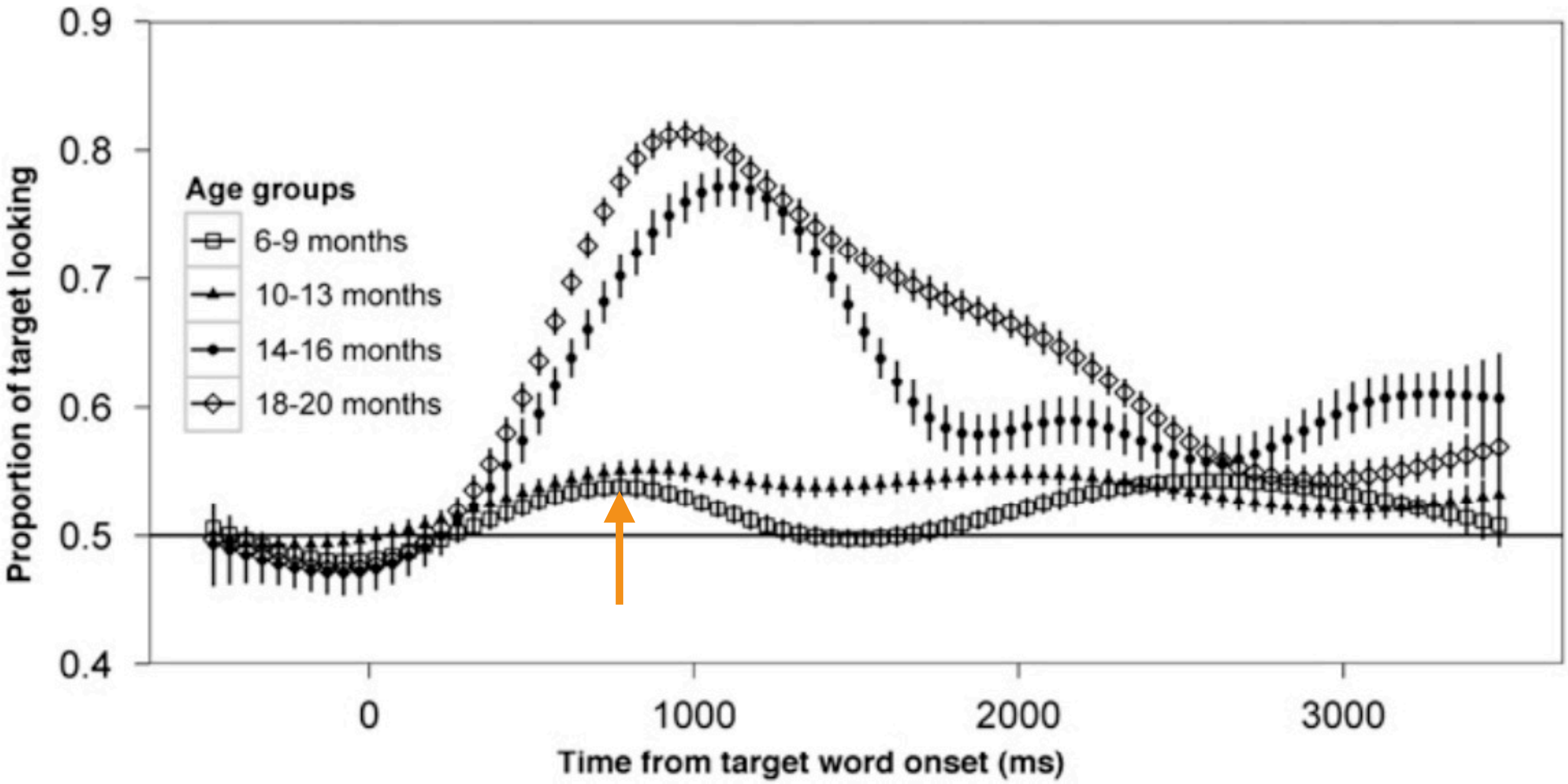



Fernald et al. (1998)

# Even at 6-9 months, infants kind of know the meanings of a few common nouns

Body words (e.g, eyes, foot…)

Food words (apple, spoon…)




(Bergelson & Swingley, 2012; Tincoff & Jusczyk, 1999)

# Early-learned words are part of an interconnected semantic network

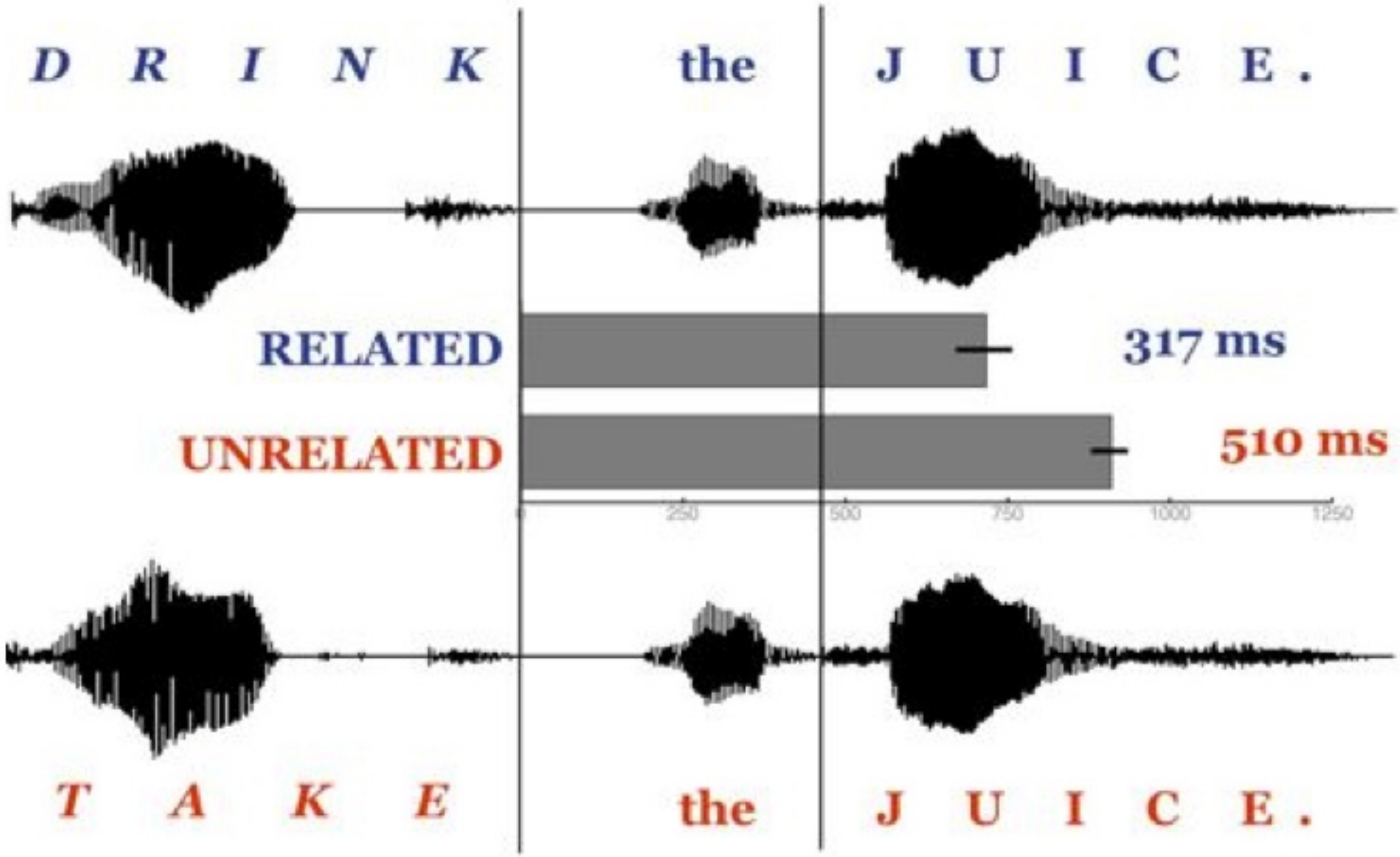


6-month-old infants are better at recognizing ‘milk’
if it’s next to an unrelated object (foot) vs. a related object (juice).

From the moment one can detect word recognition,
infants already know something about relations between words.

(Bergelson & Aslin, 2017)

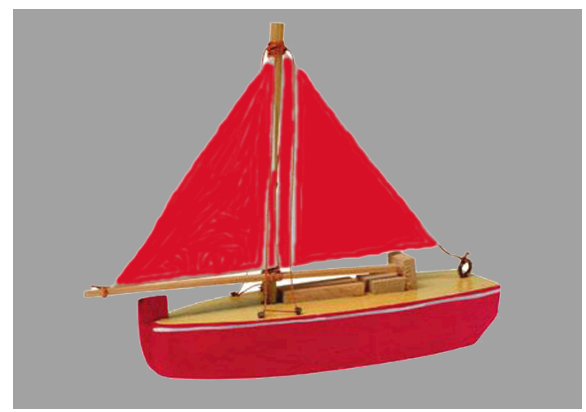
# Predicting what words come next




Toddlers can use a semantically informative verb (“drink”) to more rapidly identify a drinkable referent.

Look at the b-

Find the blue shoe



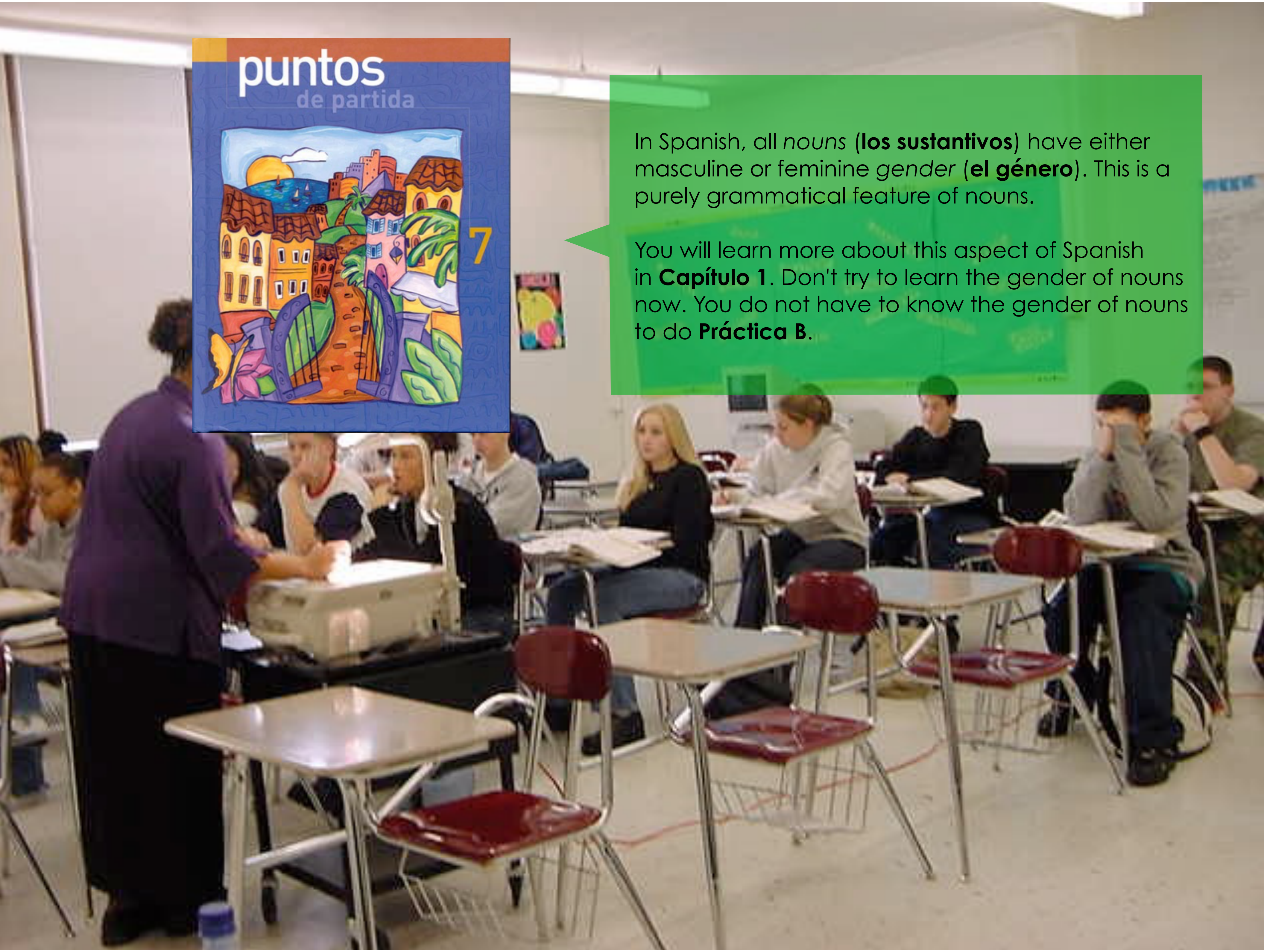
Vamos a jugar con la pelota

Ponla ahí

Ten, mira la pelota

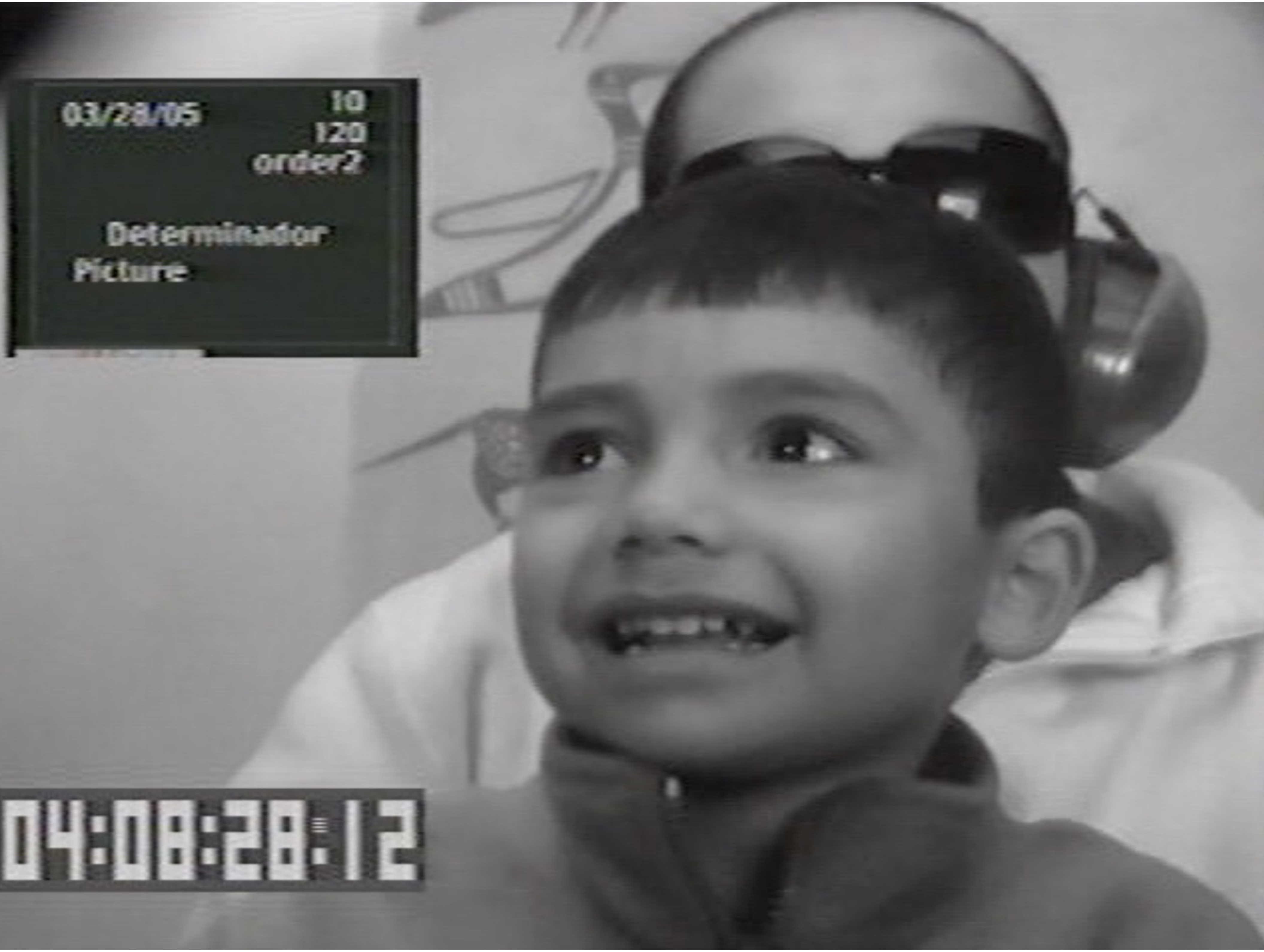
Sí, la pelota

Avientamela



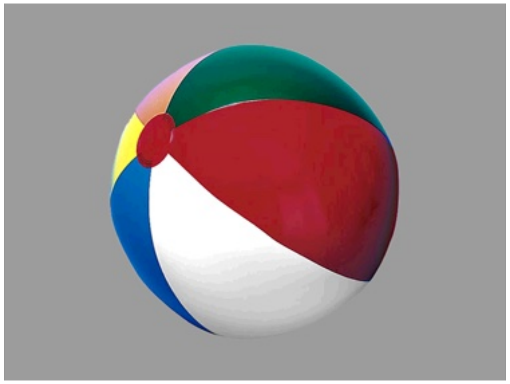

In Spanish, all *nouns* (**los sustantivos**) have either masculine or feminine *gender* (**el género**). This is a purely grammatical feature of nouns.

You will learn more about this aspect of Spanish in **Capítulo 1**. Don't try to learn the gender of nouns now. You do not have to know the gender of nouns to do **Práctica B**.


03/28/05
10
120
order2
Determinador
Picture
04:08:28:12

# ‘same-gender trials’

la galleta

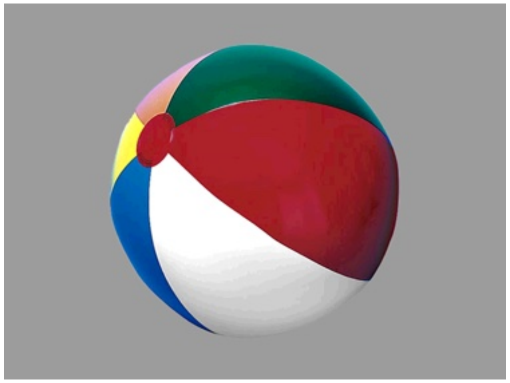


la pelota

*Encuentra la pelota*

(Lew-Williams & Fernald, 2007)

# ‘different-gender trials’



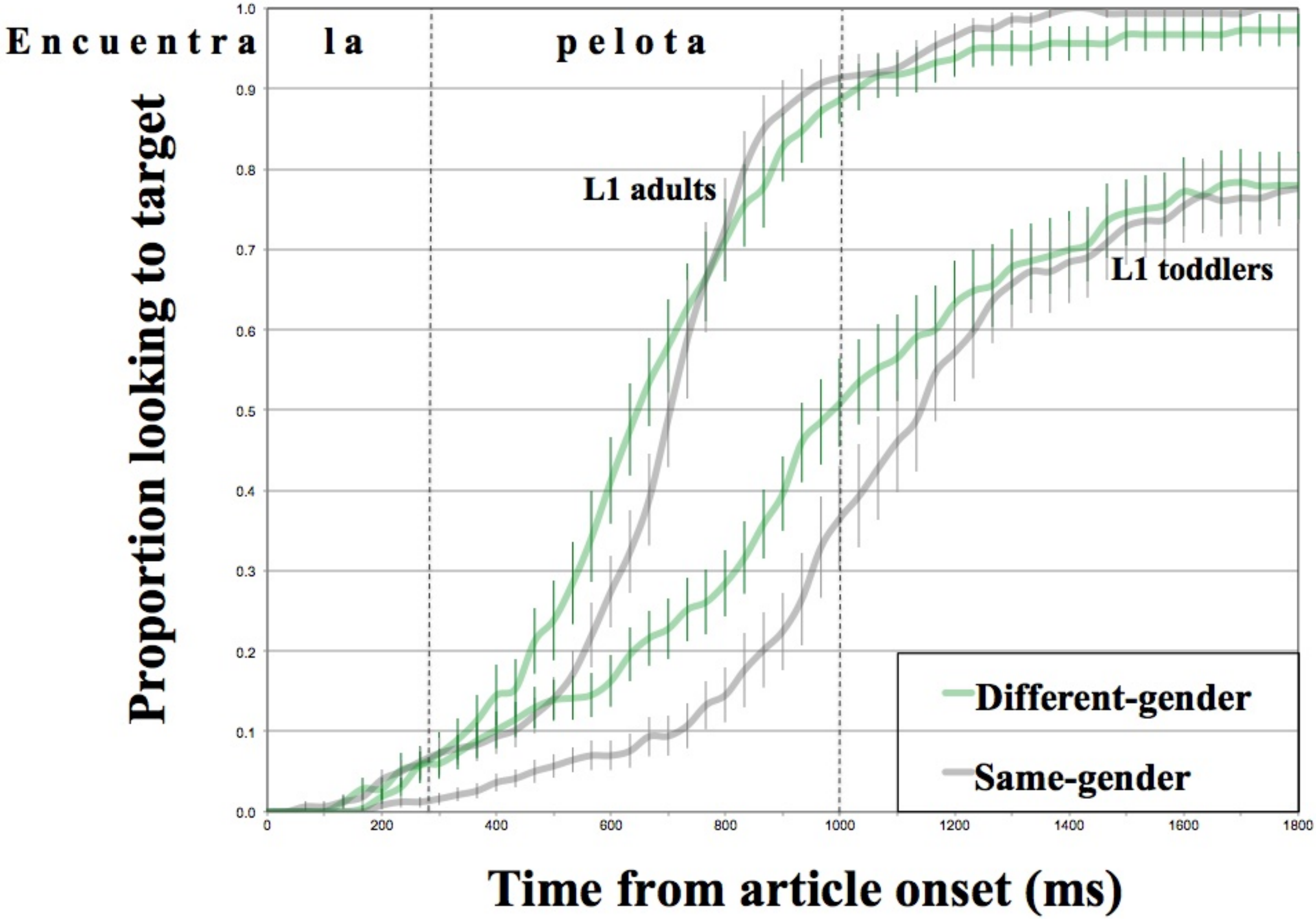
la pelota

el zapato

*Encuentra la pelota*

(Lew-Williams & Fernald, 2007)

# Results


Encuentra
la
pelota
L1 adults
L1 toddlers
Different-gender
Same-gender
Proportion looking to target
Time from article onset (ms)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
1400
1600
1800

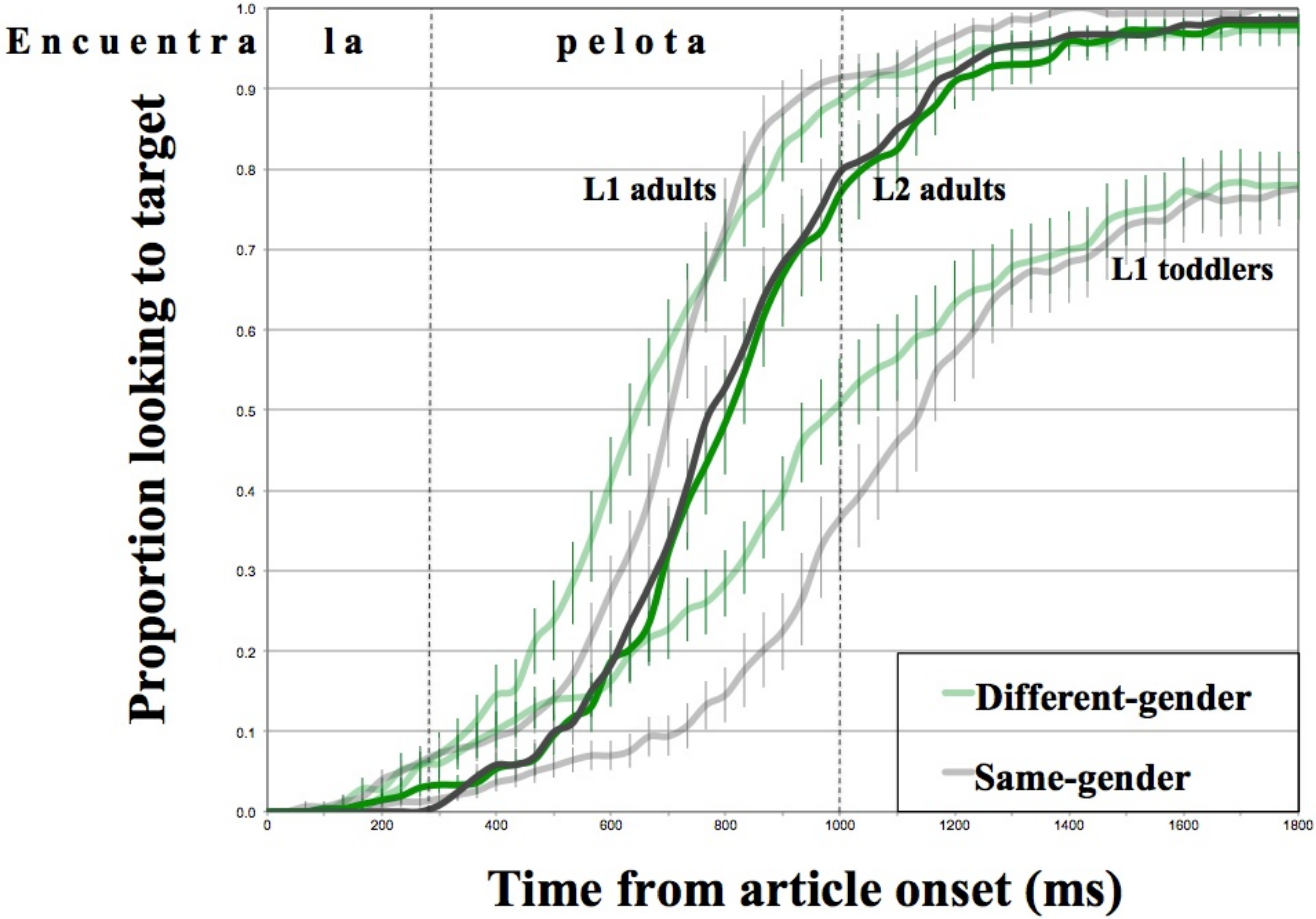

Me TALK
Pretty
ONE DAY
DAVID
SEDARIS

My confidence hit a new low when my friend Adeline told me that French children often make mistakes, but never with the [gender] of their nouns.

‘It’s just something we grow up with,’ she said. ‘We just hear the gender and think of it as part of the word.’

Tired of embarrassing myself in front of two-year-olds, I’ve started referring to everything in the plural.

# Results



Encuentra
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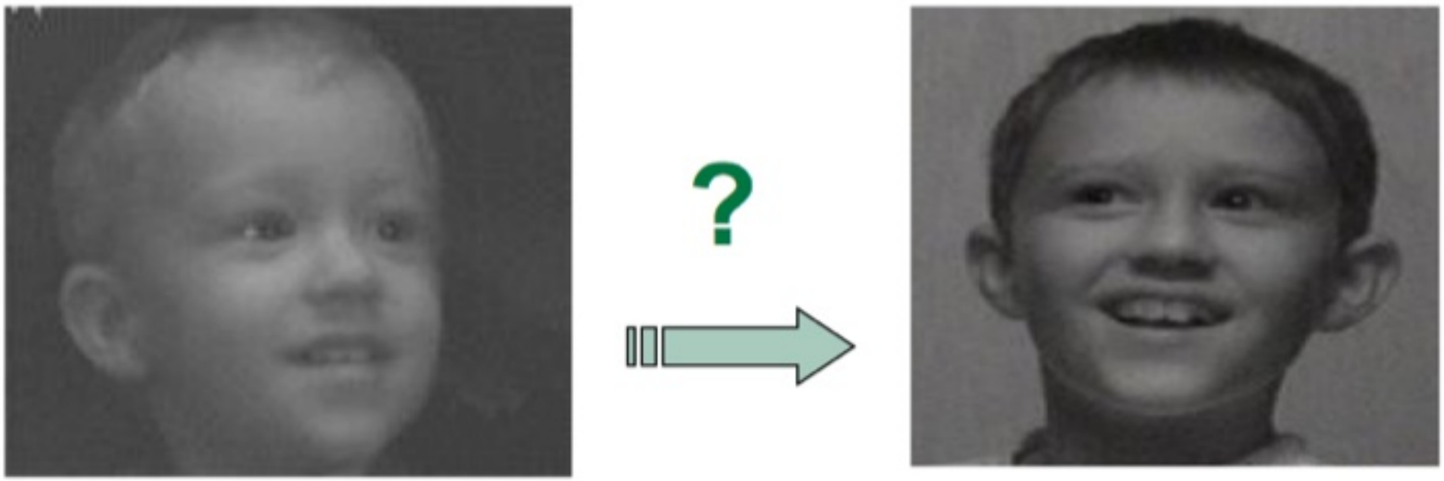
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
1400
1600
1800

# Who cares about 100 ms?

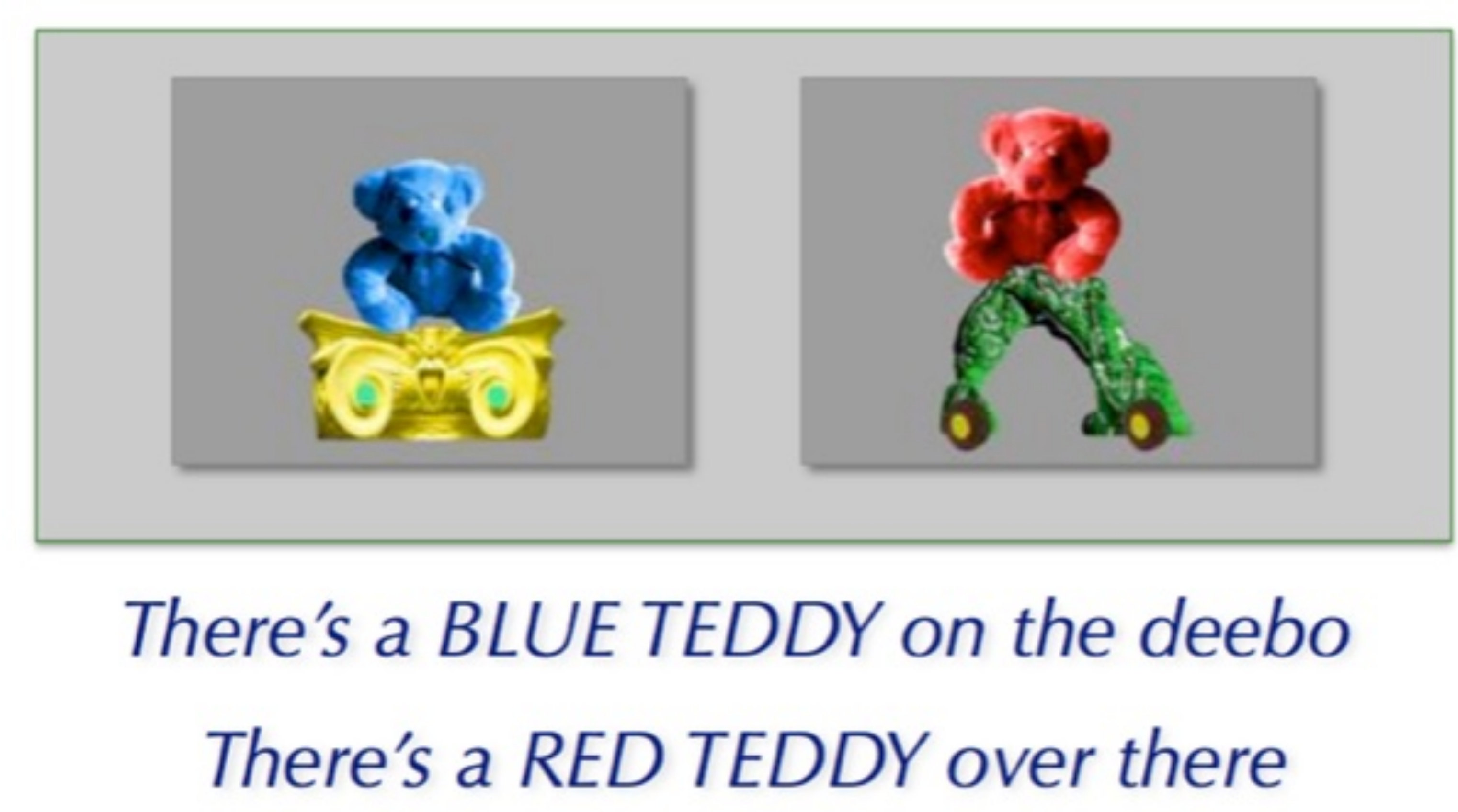
# Efficiency in processing simple sentences at age 2 predicts later language development




Vocabulary size at age 2 is correlated with language and cognitive skills at age 8, but speed of language processing doubles the predictive power!

i.e., processing is important

# Exploring links between processing efficiency and vocabulary learning




Children who are faster to respond to the familiar adjective/noun phrase were more successful in learning the new word at the end of the sentence.

michelle obama graduated from
high school in chicago

# Our predictions are often wrong…

You experience prediction errors all the time: Listening to your friends, watching tv, entering the dining hall, going up stairs…

**Prediction errors** give you an opportunity to learn something new

# Language processing as a window into atypical development

How is language learned with electric hearing (poor frequency resolution)?

Hearing-impaired 2-year-olds with cochlear implants are slower to process simple sentences.

Grieco-Calub et al. (2009)

# ‘**Late talkers**’ are a mystery

Why does vocabulary ‘bloom’ for some late talkers but drag along for others?

Factors that at least one study has found correlate with later outcomes:
(*don’t memorize these!*)

- Family history of language disorders
- Vocabulary
- Variety of babbling
- Maternal education, literacy, depression, knowledge of child development
- Birth order
- Parenting styles
- Prematurity
- Being male
- Social referencing/social engagement
- Proportion of shape-based nouns in early vocabulary
- **Real-time language processing**

(Perry, Kucker, Horst, & Samuelson, 2023)